Topic 20: Maximum ***a posteriori*** Estimation

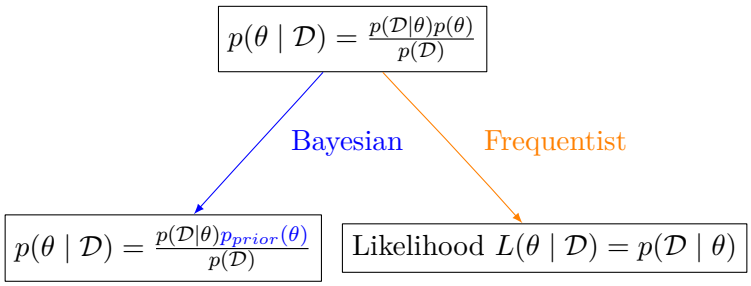
02-680: Essentials of Mathematics and Statistics

November 11, 2026

1 The Frequentist versus Bayesian Schools

Both schools of statistics start with probability. For Bayesian inference, we take θ to be a hypothesis (the parameters) and $\mathcal{D}$ some data. Different people will have different a priori beliefs — but we would still like to make useful inferences from data.

When $p(\theta)$ is known, there is no disagreement, we will all just follow Bayes' Rule (Topic 14) as written.



When the prior is *not* known, ***Bayesian*** logic requires us to develop a prior based on some information we have (intuition); while ***Frequentist*** logic uses only information in the provided data.

Bayesians and frequentists take fundamentally different approaches to this challenge. The reasons for this split are both practical (ease of implementation and computation) and philosophical (subjectivity versus objectivity and the nature of probability).

The main philosophical difference concerns the ***meaning of probability*** (recall the two perspectives from Topic 13).

Bayesians	Frequentists
the idea that probability is an abstract concept that measures a state of knowledge or a degree of belief in a given proposition	the idea that probabilities represent long-term frequencies of repeatable random experiments
Subjective interpretation	Objective interpretation
you "believe" that you will get tails 50% of the time	the relative frequency of tails goes to 1/2 as the number of flips goes to infinity
they consider a range of values each with its own probability of being true	they consider the parameter a single fixed (if unknown) value

2 Bayesians' Approach to Parameter Estimation

Let's look at the coin flip example we had in the last topic: $\mathcal{D} = X_1, X_2, \cdots, X_n$ where $X_i \sim Bernoulli(\alpha)$. We can further summarize $\mathcal{D}$ into c_H and c_T representing the counts of heads and tails respectively.

We saw last time that

$$\hat{\alpha}_{MLE} = \frac{c_H}{c_H + c_T}.$$

But this is assuming we know nothing about α ahead of time. What if we *believe* that it's 50/50, so we can add what are called ***pseudocounts*** to the input c_{H_0} and c_{T_0}. And thus compute

$$\hat{\alpha}_{MLE-PC} = \frac{c_H + c_{H_0}}{c_H + c_T + c_{H_0} + c_{T_0}}.$$

Example. Let's assume we have an experiment where we throw a coin 100 times, we want to know α, $c_H = 0$ and $c_T = 100$.

Vanilla MLE would say that

$$\hat{\alpha}_{MLE} = \frac{c_H}{c_H + c_T} = \frac{0}{0 + 100} = 0$$

But we have a small belief this is a fair coin, so let's assume we add the pseudocounts $c_{H_0} = c_{T_0} = 1$, in that case

$$\hat{\alpha}_{MLE-PC} = \frac{c_H + c_{H_0}}{c_H + c_T + c_{H_0} + c_{T_0}} = \frac{0 + 1}{0 + 100 + 1 + 1} = \frac{1}{102}.$$

If we're more confident in our prior and set $c_{H_0} = c_{T_0} = 100$ then

$$\hat{\alpha}_{MLE-PC} = \frac{c_H + c_{H_0}}{c_H + c_T + c_{H_0} + c_{T_0}} = \frac{0 + 100}{0 + 100 + 100 + 100} = \frac{100}{300} = \frac{1}{3}.$$

(We will see below that these pseudocount estimates are exactly MAP estimates under a Beta prior.)

Pseudocounts are a way of exerting your *belief*

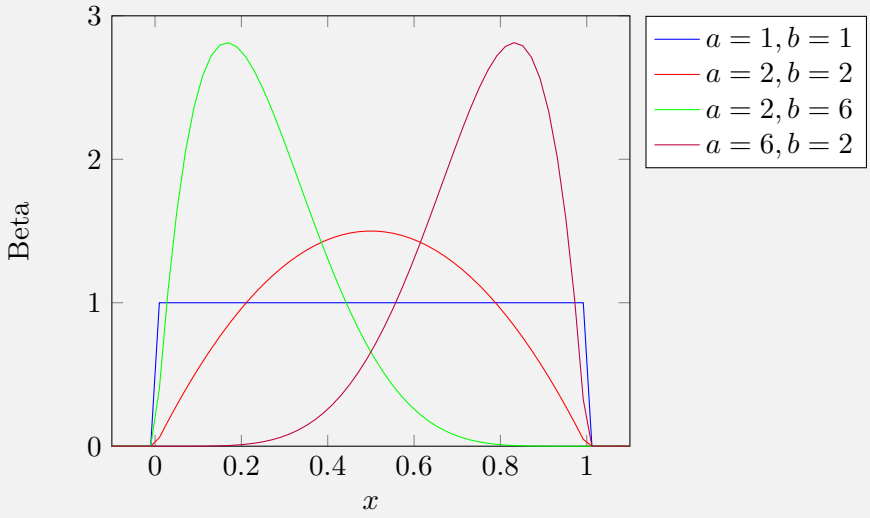
Larger pseudocounts – represent a strong prior belief
Will have a greater effect on the posterior estimate

Small pseudocounts – represent a weak prior belief
Will have a smaller effect on the posterior estimate

As the sample size goes to infinity, data will dominate the estimate.

An additional distribution: Beta

The Beta distribution visually looks like an un-balanced normal.



$$f(x; a, b) = \frac{x^{(a-1)}(1-x)^{(b-1)}}{B(a,b)}$$

where

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \text{ where } \Gamma(z) = (z-1)! \text{ for positive integers } z.$$

(For non-integer z, Γ extends the factorial smoothly, so a and b need not be integers.) When $X \sim Beta(a,b)$, $\mathbb{E}[X] = \frac{a}{a+b}$. Note $Beta(1,1)$ is the uniform distribution on $[0,1]$.

We can express a prior belief about the heads probability α (so $\theta = (\alpha)$) with a ***Beta*** distribution, that is $p(\alpha) \sim Beta(a,b)$.

What is the *posterior*, $p(\alpha \mid \mathcal{D})$?

$$\begin{aligned} p(\alpha \mid \mathcal{D}) &= \frac{p(\mathcal{D} \mid \alpha)p(\alpha)}{p(\mathcal{D})} \\ &\propto p(\mathcal{D} \mid \alpha)p(\alpha) \\ &= \alpha^{c_H}(1-\alpha)^{c_T} \cdot Beta(a,b) \\ &= \alpha^{c_H}(1-\alpha)^{c_T} \cdot \frac{\alpha^{(a-1)}(1-\alpha)^{(b-1)}}{B(a,b)} \\ &\propto \alpha^{(c_H+a-1)}(1-\alpha)^{(c_T+b-1)} \end{aligned}$$

which is the density of a $Beta(c_H + a,\ c_T + b)$ distribution. (As in Topic 19 we used the likelihood of the observed sequence. If we only recorded the counts, a binomial coefficient would appear, but it is constant in α and drops out.)

Connection to pseudocounts. The mean of a $Beta(c_H + a,\ c_T + b)$ distribution is $\frac{c_H+a}{c_H+c_T+a+b}$, so the pseudocount estimate from above with $c_{H_0} = a$ and $c_{T_0} = b$ is the posterior ***mean***. MAP instead uses the posterior ***mode***, which we derive in the example on unknown priors below.

3 Maximum *a Posteriori* (MAP) Estimation

As a reminder

$$\hat{\theta}_{MLE} = \operatorname*{argmax}_{\theta \in \Theta} p(\mathcal{D} \mid \theta).$$

On the other hand if we want to include information of our prior knowledge then we have MAP:

$$\hat{\theta}_{MAP} = \operatorname*{argmax}_{\theta \in \Theta} p(\theta \mid \mathcal{D}) = \operatorname*{argmax}_{\theta \in \Theta} p(\mathcal{D} \mid \theta) \cdot p(\theta).$$

Note that in both cases we're making a ***point estimate*** of θ. We still don't have the whole picture, we're using data to estimate our model, but MAP is ***partially Bayesian***.

If the prior is flat (uniform), $p(\theta)$ is constant and does not change the argmax, so MAP gives the same answer as MLE. That is why we could ignore $p(\theta)$ in Topic 19. A prior that is not flat, such as the pseudocounts above, pulls the estimate toward our prior belief.

3.1 Example: *Known* Prior

There are three types of coins which have different probabilities of landing heads when tossed

Type A coins are fair and have probability 0.5 of heads

Type B coins are bent and have probability 0.6 of heads

Type C coins are bent and have probability 0.9 of heads

Suppose you have a drawer containing 10 coins: 5 of type A, 3 of type B, and 2 of type C. You reach into the drawer and pick a coin at random. Without showing you the coin, it is flipped **once** and lands **tails**. What is the probability it is type A? Type B? Type C?

Hypothesis θ	Prior $p(\theta)$	Likelihood $p(\mathcal{D} \mid \theta)$	Bayes Numerator $p(\mathcal{D} \mid \theta) \cdot p(\theta)$	Posterior $p(\theta \mid \mathcal{D})$
A	0.5	0.5	0.25	≈ 0.641
B	0.3	0.4	0.12	≈ 0.308
C	0.2	0.1	0.02	≈ 0.051

Thus $\hat{\theta}_{MAP} = A$.

What if you then flip the same coin another 9 times, so including the first coin we have $c_H = 6$ and $c_T = 4$. Notice in the table below, the prior does not change.

Hypothesis θ	Prior $p(\theta)$	Likelihood $p(\mathcal{D} \mid \theta)$	Bayes Numerator $p(\mathcal{D} \mid \theta) \cdot p(\theta)$	Posterior $p(\theta \mid \mathcal{D})$
A	0.5	0.98×10^{-3}	4.88×10^{-4}	0.570
B	0.3	1.19×10^{-3}	3.58×10^{-4}	0.418
C	0.2	0.05×10^{-3}	0.11×10^{-4}	0.012

In this case, it's still true that $\hat{\theta}_{MAP} = A$, but $\hat{\theta}_{MLE} = B$.

3.2 *Unknown* Prior

Assume we have a similar scenario but this time there is no list of hypotheses with known prior probabilities (as in the coin drawer). Instead we choose a prior ***distribution*** for the parameter (see conjugate priors below), and follow a similar procedure to that for MLE: take the zero point of the log of the posterior.

$$\frac{d}{d\theta} \ln\big[p(\mathcal{D} \mid \theta)\, p(\theta)\big] = 0$$

Example: Bernoulli with a Beta prior. Let $X_i \sim Bernoulli(\alpha)$ with c_H heads and c_T tails, and a prior $\alpha \sim Beta(a, b)$. From above, the posterior is proportional to

$\alpha^{(c_H+a-1)}(1-\alpha)^{(c_T+b-1)}$. Following the recipe from Topic 19, take the log and find the zero of the derivative:

$$\begin{aligned}\frac{d}{d\alpha}\ln\left[\alpha^{(c_H+a-1)}(1-\alpha)^{(c_T+b-1)}\right] &= \frac{d}{d\alpha}\left[(c_H+a-1)\ln\alpha+(c_T+b-1)\ln(1-\alpha)\right] \\ &= \frac{c_H+a-1}{\alpha}-\frac{c_T+b-1}{1-\alpha}=0.\end{aligned}$$

Solving gives

$$\hat{\alpha}_{MAP}=\frac{c_H+a-1}{c_H+c_T+a+b-2}.$$

This is the Bernoulli MLE from Topic 19 with $a-1$ extra heads and $b-1$ extra tails, so it is exactly the pseudocount estimate from above with pseudocounts $c_{H_0}=a-1$ and $c_{T_0}=b-1$. For the coin with $c_H=0$ and $c_T=100$: a $Beta(2,2)$ prior (pseudocounts of 1) gives $\hat{\alpha}_{MAP}=\frac{1}{102}$, while the flat $Beta(1,1)$ prior gives $\hat{\alpha}_{MAP}=0$, the same as the MLE.

Conjugate Priors

Definition: If the prior and the posterior belong to the same parametric family, then the prior is said to be conjugate for the likelihood. Parameters of the prior distribution are often called "hyperparameters".
Example: a, b in $Beta(a,b)$.

How do you set the hyperparameters?

- Prior knowledge (from experienced domain-specific experts)
- If no prior knowledge, use "uninformative" prior (e.g., $Beta(1,1)$, which is flat, so MAP equals MLE)

Some common conjugate prior sets are in the table below:

Prior	Likelihood	Posterior
Beta	Bernoulli	Beta
Beta	Binomial	Beta
Gamma	Poisson	Gamma
Normal	Normal	Normal

The Bernoulli, Normal, and Poisson likelihoods are the ones whose MLEs we derived in Topic 19. (We do not define the Gamma distribution here; it is the standard prior for a Poisson rate λ.)

Useful References

Wasserman. "All of Statistics: A Concise Course in Statistical Inference" §11 Degroot and Schervish. "Probability and Statistics" §§7.1-7.4